Term Test 1 solutions

Calculus (University of Toronto)



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University of Toronto
Faculty of Arts and Science

MAT137Y Test 1

X. Cui, I. Gaiur, T. Kojar, L. Salvetti Martinez, T. Xian, J. Zhang, X. Zhang

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Duration: 110 minutes

No Aids Permitted

Sample Solutions

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1. For each of the following questions, only your final answer will be graded.

No justification is necessary.

Evaluate the following limits. If a limit is ∞ or $-\infty$, clearly state this. If the limit does not exist and it's not ∞ or $-\infty$, put "DNE" in the box.

(1a) (2 points) Given the function $f(x) = \begin{cases} 7 - 4x & x < 1 \\ x^2 + 3 & x \geq 1 \end{cases}$.

- Find $\lim_{x \to -2} f(x)$.

Final Answer

- Find $\lim_{x \to 1} f(x)$.

Final Answer

(1b) (1 point) $\lim_{x \to \pi^-} \ln\left(\frac{1}{\sin x}\right)$

Final Answer

(1c) (1 point) $\lim_{x \to 4} \frac{\sqrt{x} - 2}{x - 4}$

Final Answer

(1d) (1 point) $\lim_{x \to +\infty} \frac{\cos(2x) + 2}{x^2 + x + 1}$

Final Answer

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For each of the following questions, only your final answer will be graded.

No justification is necessary.

2. (2 points) Let $a, L \in \mathbb{R}$. Let f be a function defined on $\mathbb{R}$. Write the complete $\varepsilon - \delta$ formal definition for "$\lim_{x \to a^+} f(x) \neq L$".

Final Answer

[redacted]

3. (2 points) Let f be a function defined on $\mathbb{R}$. Write the complete $\varepsilon - \delta$ formal definition for "$f(x)$ is continuous at a".

Final Answer

[redacted]

4. (3 points) Classify the following statements as **TRUE**, **FALSE**.

   (a) $\forall x \in \mathbb{Z}, \exists y \in \mathbb{Q}$ such that $xy = 1$.

   ○ True ○ False

   (b) $5 \in \left\{x \in \mathbb{R} : \exists y \in \mathbb{Z} \text{ } s.t. \text{ } y^2 > 1 \text{ } and \text{ } y < x\right\}$.

   ○ True ○ False

   (c) $\forall x \in \mathbb{Q}, x > 0 \implies x \geq 0$

   ○ True ○ False

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For each of the following questions, only your final answer will be graded.

No justification is necessary.

5. (5a) (1 point) Write down the set of rational numbers using set builder notation.

Final Answer

[redacted]

(5b) (1 point) Express the following set $\{x \in [0,1] : \exists y \in [-1,1],\ s.t.\ x > y^2\}$ in interval notation.

Final Answer

[redacted]

(5c) (1 point) Find a function f with domain $\mathbb{R}$ such that

- f is continuous on $\mathbb{R}$, except at 0.
- $\lim_{x \to 0^+} f(x) \neq \lim_{x \to 0^-} f(x)$ and $f(0) = 1$.

It is okay to use piece-wise-defined functions.

Final Answer

[redacted]

(5d) (1 point) Find a pair of two functions f and g defined on $\mathbb{R}$ such that $\lim_{x \to 1} f(x) = 0$ and $\lim_{x \to 1} f(x)g(x) \neq 0$. It is okay to use piece-wise-defined functions.

Final Answer

[redacted]

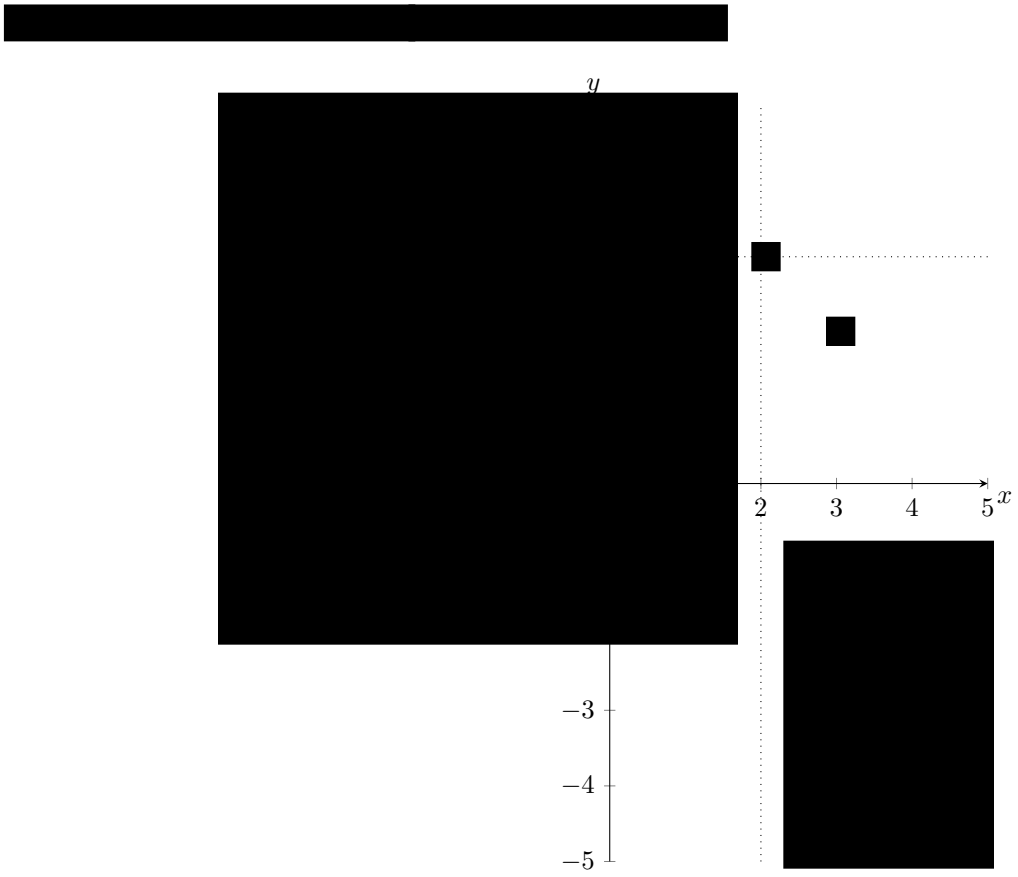
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6. (6 points) Sketch the graph of a function f defined on $\mathbb{R}$ which satisfies the following properties at once.

- $\lim_{x\to 0} f(x) = -2$
- $\lim_{x\to 0} f(f(x)) = 1$
- $\lim_{x\to 0^-} f(\frac{1}{x}) = 3$
- $\lim_{x\to 2^+} f(x) = -\infty$
- $\lim_{x\to 2}[f(x)]^2 = \infty$
- $\lim_{x\to 3} f(x) = -1, f(3) = 2$



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7. Let $f(x)$ be a function defined on $\mathbb{R}$ such that $\lim_{x\to\infty} f(x) = 2$.

(7a) (2 points) Write the formal definition of $\lim_{x\to\infty} f(x) = 2$.

[redacted]

(7b) (5 points) Use the formal ε-δ definition of the limit to prove

$$\lim_{x\to\infty} 4f(3x+1) = 8.$$

Do not use any limit laws or any other theorems.

[redacted]

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8. (5 points) Define $f(n) = 1^2 + 2^2 + 3^2 + \cdots + n^2$ where n is any positive integer. Prove the following statement by using math induction.

$$P(n) : f(n) \leq n^3$$

is true for all positive integer n.



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9. Let A, B be any subsets of $\mathbb{R}$ and non-empty.
We say A is *calm* if

$$\exists a, b \in A, \ s.t. \ \forall x \in A, a \leq x \leq b$$

We say B is *happy* if

$$\exists a \in B, \ s.t. \ \forall x \in B, x \geq a$$

Determine if the following statements are true or false. If a claim is true, prove it. If a claim is false, provide a counterexample and a justification of how the counterexample shows the claim is false.

(9a) (3 points) If a non-empty set B is happy, then it is calm.

○ True ○ False

[redacted]

(9b) (3 points) If a non-empty set A is calm and a non-empty set B is happy, $A \neq B$, then $A \cup B$ is happy.

○ True ○ False

[redacted]

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